Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

John would have reduced the time it took him to drive from his home to a certain store by $\frac{1}{3}$ if he had increased his average speed by 15 miles per hour. What was John's actual average speed, in miles per hour, when he drove from his home to the store?

A. 25

B. 30

C. 40

D. 45

E. 50

•

Which of the following inequalities has a solution set that, when graphed on a number line, is a single line segment of finite length?

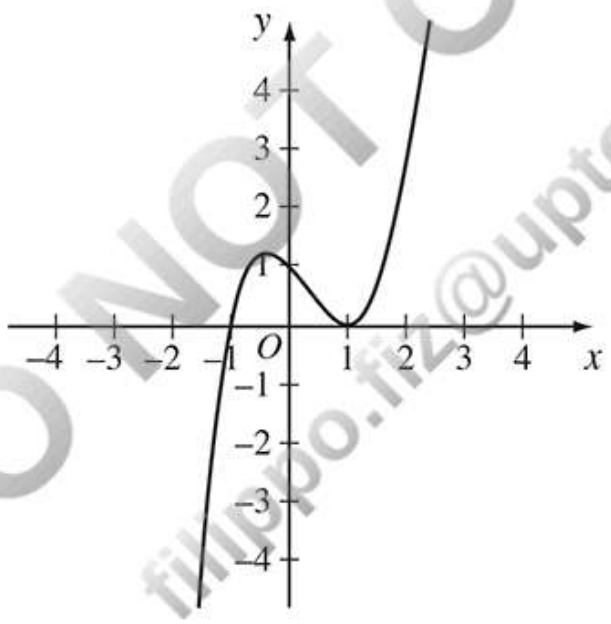
A. $x^4 \geq 1$

B. $x^3 \leq 27$

C. $x^2 \geq 16$

D. $2 \leq |x| \leq 5$

E. $2 \leq 3x + 4 \leq 6$



DO NOT COPY
filippo.fiz@uptoten.it

•

The figure shows the graph of $y = \left(x+1\right)(x-1)^2$ in the xy-plane. At how many points does the graph of $y = \left(x+1\right)(x-1)^2 + 2$ intercept the x-axis?

A. None

B. One

C. Two

D. Three

E. Four

Questions 177 to 272 — Difficulty: Hard

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Questions 177 to 272 — Difficulty: Hard

177. The cost to rent a small bus for a trip is x dollars, which is to be shared equally among the people taking the trip. If 10 people take the trip rather than 16, how many more dollars, in terms of x, will it cost per person?

A. $\frac{x}{6}$

B. $\frac{x}{10}$

C. $\frac{x}{16}$

D. $\frac{3x}{40}$

E. $\frac{3x}{80}$

178. Clarissa will create her summer reading list by randomly choosing 4 books from the 10 books approved for summer reading. She will list the books in the order in which they are chosen. How many different lists are possible?

A. 6

B. 40

C. 210

D. 5,040

E. 151,200

179. If n is a positive integer and the product of all the integers from 1 to n, inclusive, is divisible by 990, what is the least possible value of n?

A. 8

B. 9

C. 10

D. 11

E. 12

180. The probability that event M will NOT occur is 0.8 and the probability that event R will NOT occur is 0.6. If events M and R CANNOT both occur, which of the following is the probability that either event M or event R will occur?

A. $\frac{1}{5}$

B. $\frac{2}{5}$

C. $\frac{3}{5}$

D. $\frac{4}{5}$

E. $\frac{12}{25}$

181. The total cost for Company X to produce a batch of tools is \$10,000 plus \$3 per tool. Each tool sells for \$8.The gross profit earned from producing and selling these tools is the total income from sales minus the total production cost. If a batch of 20,000 tools is produced and sold, then Company X's gross profit per tool is

A. \$3.00

B. \$3.75

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

B. $3.75

C. $4.50

D. $5.00

E. $5.50

182. If Q is an odd number and the median of Q consecutive integers is 120, what is the largest of these integers?

A. $\frac{Q-1}{2}+120$

B. $\frac{Q}{2}+199$

C. $\frac{Q}{2}+120$

D. $\frac{Q+119}{2}$

E. $\frac{Q+120}{2}$

183. If there are fewer than 8 zeros between the decimal point and the first nonzero digit in the decimal expansion of $\left(\frac{t}{1,000}\right)^4$, which of the following numbers could be the value of t?

I. 3

II. 5

III. 9

A. None

B. I only

C. II only

D. III only

E. II and III

184. A three-digit code for certain locks uses the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 according to the following constraints. The first digit cannot be 0 or 1, the second digit must be 0 or 1, and the second and third digits cannot both be 0 in the same code. How many different codes are possible?

A. 144

B. 152

C. 160

D. 168

E. 176

185. Jackie has two solutions that are 2 percent sulfuric acid and 12 percent sulfuric acid by volume, respectively. If these solutions are mixed in appropriate quantities to produce 60 liters of a solution that is 5 percent sulfuric acid, approximately how many liters of the 2 percent solution will be required?

A. 18

B. 20

C. 24

D. 36

E. 42

186. If Jake loses 8 pounds, he will weigh twice as much as his sister. Together they now weigh 278 pounds. What is

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

186. If Jake loses 8 pounds, he will weigh twice as much as his sister. Together they now weigh 278 pounds. What is Jake's present weight, in pounds?

A. 131

B. 135

C. 139

D. 147

E. 188

187. For each student in a certain class, a teacher adjusted the student's test score using the formula $y = 0.8x + 20$, where x is the student's original test score and y is the student's adjusted test score. If the standard deviation of the original test scores of the students in the class was 20, what was the standard deviation of the adjusted test scores of the students in the class?

A. 12

B. 16

C. 28

D. 36

E. 40

188. Last year 26 members of a certain club traveled to England, 26 members traveled to France, and 32 members traveled to Italy. Last year no members of the club traveled to both England and France, 6 members traveled to both England and Italy, and 11 members traveled to both France and Italy. How many members of the club traveled to at least one of these three countries last year?

A. 52

B. 67

C. 71

D. 73

E. 79

189. A store reported total sales of $385 million for February of this year. If the total sales for the same month last year was $320 million, approximately what was the percent increase in sales?

A. 2%

B. 17%

C. 20%

D. 65%

E. 83%

190. When positive integer xv is divided by positive integer y, the remainder is 9. If $\frac{x}{y} = 96.12$, what is the value of y?

A. 96

B. 75

C. 48

D. 25

E. 12

191. If $x(2x+1) = 0$ and $\left(x + \frac{1}{2}\right)\left(2x - 3\right) = 0$, then $x =$

A. -3

B. 1

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

B. $-\frac{1}{2}$

C. 0

D. $\frac{1}{2}$

E. $\frac{3}{2}$

192. A certain experimental mathematics program was tried out in 2 classes in each of 32 elementary schools and involved 37 teachers. Each of the classes had 1 teacher and each of the teachers taught at least 1, but not more than 3, of the classes. If the number of teachers who taught 3 classes is n, then the least and greatest possible values of n, respectively, are

A. 0 and 13

B. 0 and 14

C. 1 and 10

D. 1 and 9

E. 2 and 8

193. For the positive numbers, n, $n+1$, $n+2$, $n+4$, and $n+8$, the mean is how much greater than the median?

A. 0

B. 1

C. $n+1$

D. $n+2$

E. $n+3$

194. The present ratio of students to teachers at a certain school is 30 to 1. If the student enrollment were to increase by 50 students and the number of teachers were to increase by 5, the ratio of students to teachers would then be 25 to 1. What is the present number of teachers?

A. 5

B. 8

C. 10

D. 12

E. 15

195. What is the smallest integer n for which $25^n > 5^{12}$?

A. 6

B. 7

C. 8

D. 9

E. 10

196. Sixty percent of the members of a study group are women, and 45 percent of those women are lawyers. If one member of the study group is to be selected at random, what is the probability that the member selected is a woman lawyer?

A. 0.10

B. 0.15

C. 0.27

D. 0.33

E. 0.45

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

E. 0.45

197. Each year for 4 years, a farmer increased the number of trees in a certain orchard by $\frac{1}{4}$ of the number of trees in the orchard the preceding year. If all of the trees thrived and there were 6,250 trees in the orchard at the end of the 4-year period, how many trees were in the orchard at the beginning of the 4-year period?

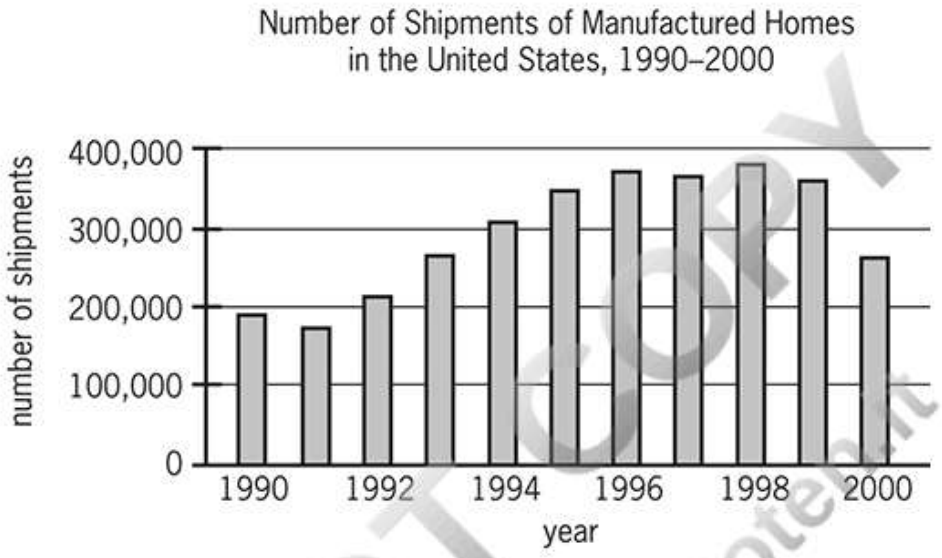
A. 1,250

B. 1,563

C. 2,250

D. 2,560

E. 2,752



198. According to the chart shown, which of the following is closest to the median annual number of shipments of manufactured homes in the United States for the years from 1990 to 2000, inclusive?

A. 250,000

B. 280,000

C. 310,000

D. 325,000

E. 340,000

199. For the positive integers a, b, and k, $a^k \mid\mid b$ means that a^k is a divisor of b, but a^{k+1} is not a divisor of b. If k is a positive integer and $2^k \mid\mid 72$, then k is equal to

A. 2

B. 3

C. 4

D. 8

E. 18

200. A certain characteristic in a large population has a distribution that is symmetric about the mean m.If 68 percent of the distribution lies within one standard deviation d of the mean, what percent of the distribution is less than $m + d$?

A. 16%

B. 32%

C. 48%

D. 84%

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

D. 84%

E. 92%

201. Four extra-large sandwiches of exactly the same size were ordered for m students, where $m > 4$. Three of the sandwiches were evenly divided among the students. Since 4 students did not want any of the fourth sandwich, it was evenly divided among the remaining students. If Carol ate one piece from each of the four sandwiches, the amount of sandwich that she ate would be what fraction of a whole extra-large sandwich?

A. $\frac{m+4}{m(m-4)}$

B. $\frac{2m-4}{m(m-4)}$

C. $\frac{4m-4}{m(m-4)}$

D. $\frac{4m-8}{m(m-4)}$

E. $\frac{4m-12}{m(m-4)}$

202. Which of the following equations has $1+\sqrt{2}$ as one of its roots?

A. $x^2+2x-1=0$

B. $x^2-2x+1=0$

C. $x^2+2x+1=0$

D. $x^2-2x-1=0$

E. $x^2-x-1=0$

203. In Country C, the unemployment rate among construction workers dropped from 16 percent on September 1, 1992, to 9 percent on September 1, 1996. If the number of construction workers was 20 percent greater on September 1, 1996, than on September 1, 1992, what was the approximate percent change in the number of unemployed construction workers over this period?

A. 50% decrease

B. 30% decrease

C. 15% decrease

D. 30% increase

E. 55% increase

204. In a box of 12 pens, a total of 3 are defective. If a customer buys 2 pens selected at random from the box, what is the probability that neither pen will be defective?

A. $\frac{1}{6}$

B. $\frac{2}{9}$

C. $\frac{6}{11}$

D. $\frac{9}{16}$

E. $\frac{3}{4}$

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

205. At a certain fruit stand, the price of each apple is 40 cents and the price of each orange is 60 cents. Mary selects a total of 10 apples and oranges from the fruit stand, and the average (arithmetic mean) price of the 10 pieces of fruit is 56 cents. How many oranges must Mary put back so that the average price of the pieces of fruit that she keeps is 52 cents?

A. 1

B. 2

C. 3

D. 4

E. 5

206. A pharmaceutical company received $3 million in royalties on the first $20 million in sales of the generic equivalent of one of its products and then $9 million in royalties on the next $108 million in sales. By approximately what percent did the ratio of royalties to sales decrease from the first $20 million in sales to the next $108 million in sales?

A. 8%

B. 15%

C. 45%

D. 52%

E. 56%

Times at Which the Door Opened from 8:00 to 10:00

8:00	8:06	8:30	9:05
8:03	8:10	8:31	9:11
8:04	8:18	8:54	9:29
8:04	8:19	8:57	9:31

207. The light in a restroom operates with a 15-minute timer that is reset every time the door opens as a person goes in or out of the room. Thus, after someone enters or exits the room, the light remains on for only 15 minutes unless the door opens again and resets the timer for another 15 minutes. If the times listed above are the times at which the door opened from 8:00 to 10:00, approximately how many minutes during this two-hour period was the light off?

A. 10

B. 25

C. 35

D. 40

E. 70

208. If p is the product of the integers from 1 to 30, inclusive, what is the greatest integer k for which 3^k is a factor of p?

A. 10

B. 12

C. 14

D. 16

E. 18

209. If $n = 3^8 - 2^8$, which of the following is NOT a factor of n?

A. 97

B. 65

C. 35

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

C. 35

D. 13

E. 5

210. Club X has more than 10 but fewer than 40 members. Sometimes the members sit at tables with 3 members at one table and 4 members at each of the other tables,and sometimes they sit at tables with 3 members at one table and 5 members at each of the other tables. If they sit at tables with 6 members at each table except one and fewer than 6 members at that one table, how many members will be at the table that has fewer than 6 members?

A. 1

B. 2

C. 3

D. 4

E. 5

211. In order to complete a reading assignment on time, Terry planned to read 90 pages per day. However, she read only 75 pages per day at first, leaving 690 pages to be read during the last 6 days before the assignment was to be completed. How many days in all did Terry have to complete the assignment on time?

A. 15

B. 16

C. 25

D. 40

E. 46

212. If $s > 0$ and $\sqrt{\frac{r}{s}} = s$, what is r in terms of s?

A. $\frac{1}{s}$

B. $\sqrt{s}$

C. $s\sqrt{s}$

D. s^3

E. $s^2 - s$

213. If $3 < x < 100$, for how many values of x is $\frac{x}{3}$ the square of a prime number?

A. Two

B. Three

C. Four

D. Five

E. Nine

214. A researcher plans to identify each participant in a certain medical experiment with a code consisting of either a single letter or a pair of distinct letters written in alphabetical order. What is the least number of letters that can be used if there are 12 participants, and each participant is to receive a different code?

A. 4

B. 5

C. 6

D. 7

E. 8

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

215. An object thrown directly upward is at a height of h feet after t seconds, where $h = -16(t-3)^2 + 150$. At what height, in feet, is the object 2 seconds after it reaches its maximum height?

A. 6

B. 86

C. 134

D. 150

E. 166

216. Which of the following is equivalent to the pair of inequalities $x + 6 > 10$ and $x - 3 \leq 5$?

A. $2 \leq x < 16$

B. $2 \leq x < 4$

C. $2 < x \leq 8$

D. $4 < x \leq 8$

E. $4 \leq x < 16$

217. David has d books, which is 3 times as many as Jeff and $\frac{1}{2}$ as many as Paula. How many books do the three of them have altogether, in terms of d?

A. $\frac{5}{6}d$

B. $\frac{7}{3}d$

C. $\frac{10}{3}d$

D. $\frac{7}{2}d$

E. $\frac{9}{2}d$

218. There are 8 teams in a certain league and each team plays each of the other teams exactly once. If each game is played by 2 teams, what is the total number of games played?

A. 15

B. 16

C. 28

D. 56

E. 64

219. At his regular hourly rate, Don had estimated the labor cost of a repair job as \$336 and he was paid that amount. However, the job took 4 hours longer than he had estimated and, consequently, he earned \$2 per hour less than his regular hourly rate. What was the time Don had estimated for the job, in hours?

A. 28

B. 24

C. 16

D. 14

E. 12

220. If $\frac{p}{q} < 1$, and p and q are positive integers, which of the following must be greater than 1?

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

q

A. $\sqrt{\frac{p}{q}}$

B. $\frac{p}{q^2}$

C. $\frac{p}{2q}$

D. $\frac{q}{p^2}$

E. $\frac{q}{p}$

221. To mail a package, the rate is x cents for the first pound and y cents for each additional pound, where $x > y$. Two packages weighing 3 pounds and 5 pounds, respectively, can be mailed separately or combined as one package. Which method is cheaper, and how much money is saved?

A. Combined, with a savings of $x - y$ cents

B. Combined, with a savings of $y - x$ cents

C. Combined, with a savings of x cents

D. Separately, with a savings of $x - y$ cents

E. Separately, with a savings of y cents

222. If money is invested at r percent interest, compounded annually, the amount of the investment will double in approximately $\frac{70}{r}$ years. If Pat's parents invested \$5,000 in a long-term bond that pays 8 percent interest, compounded annually, what will be the approximate total amount of the investment 18 years later, when Pat is ready for college?

A. \$20,000

B. \$15,000

C. \$12,000

D. \$10,000

E. \$9,000

223. On a recent trip, Cindy drove her car 290 miles, rounded to the nearest 10 miles, and used 12 gallons of gasoline, rounded to the nearest gallon. The actual number of miles per gallon that Cindy's car got on this trip must have been between

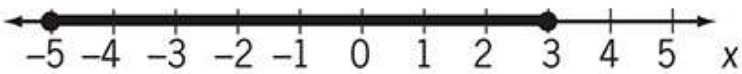
A. $\frac{290}{12.5}$ and $\frac{290}{11.5}$

B. $\frac{295}{12}$ and $\frac{285}{11.5}$

C. $\frac{285}{12}$ and $\frac{295}{12}$

D. $\frac{285}{12.5}$ and $\frac{295}{11.5}$

E. $\frac{295}{12.5}$ and $\frac{285}{11.5}$



DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

224. Which of the following inequalities is an algebraic expression for the shaded part of the number line above?

A. $|x| \leq 3$

B. $|x| \leq 5$

C. $|x-2| \leq 3$

D. $|x-1| \leq 4$

E. $|x+1| \leq 4$

225. In a small snack shop, the average (arithmetic mean) revenue was $400 per day over a 10-day period. During this period, if the average daily revenue was $360 for the first 6 days, what was the average daily revenue for the last 4 days?

A. $420

B. $440

C. $450

D. $460

E. $480

226. If y is the smallest positive integer such that 3,150 multiplied by y is the square of an integer, then y must be

A. 2

B. 5

C. 6

D. 7

E. 14

227. If $[x]$ is the greatest integer less than or equal to x, what is the value of [–1.6] + [3.4] + [2.7]?

A. 3

B. 4

C. 5

D. 6

E. 7

228. In the first week of the year, Nancy saved $1. In each of the next 51 weeks, she saved $1 more than she had saved in the previous week. What was the total amount that Nancy saved during the 52 weeks?

A. $1,326

B. $1,352

C. $1,378

D. $2,652

E. $2,756

229. In a certain sequence, the term x_n is given by the formula $x_n = 2x_{n-1} - \frac{1}{2}(x_{n-2})$ for all $n \geq 2$. If $x_0 = 3$ and $x_1 = 2$, what is the value of x_3?

A. 2.5

B. 3.125

C. 4

D. 5

E. 6.75

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

230. During a trip, Francine traveled x percent of the total distance at an average speed of 40 miles per hour and the rest of the distance at an average speed of 60 miles per hour. In terms of x, what was Francine's average speed for the entire trip?

A. $\frac{180-x}{2}$

B. $\frac{x+60}{4}$

C. $\frac{300-x}{5}$

D. $\frac{600}{115-x}$

E. $\frac{12,000}{x+200}$

231. If $n=(33)^{43}+(43)^{33}$, what is the units digit of n?

A. 0

B. 2

C. 4

D. 6

E. 8

232. Team A and Team B are competing against each other in a game of tug-of-war. Team A, consisting of 3 males and 3 females, decides to line up male, female, male, female, male, female. The lineup that Team A chooses will be one of how many different possible lineups?

A. 9

B. 12

C. 15

D. 36

E. 720

233. If $d=\frac{1}{2^3\times 5^7}$ is expressed as a terminating decimal, how many nonzero digits will d have?

A. One

B. Two

C. Three

D. Seven

E. Ten

234. For any positive integer n, the sum of the first n positive integers equals $\frac{n(n+1)}{2}$. What is the sum of all the even integers between 99 and 301?

A. 10,100

B. 20,200

C. 22,650

D. 40,200

E. 45,150

235. November 16, 2001, was a Friday. If each of the years 2004, 2008, and 2012 had 366 days, and the remaining

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

235. November 16, 2001, was a Friday. If each of the years 2004, 2008, and 2012 had 366 days, and the remaining years from 2001 through 2014 had 365 days, what day of the week was November 16, 2014?

A. Sunday

B. Monday

C. Tuesday

D. Wednesday

E. Thursday

236. How many prime numbers between 1 and 100 are factors of 7,150?

A. One

B. Two

C. Three

D. Four

E. Five

237. A sequence of numbers $a_1, a_2, a_3, \ldots$ is defined as follows: $a_1 = 3$, $a_2 = 5$, and every term in the sequence after a_2 is the product of all terms in the sequence preceding it, e.g., $a_3 = (a_1)(a_2)$ and $a_4 = (a_1)(a_2)(a_3)$. If $a_n = t$ and $n > 2$, what is the value of a_{n+2} in terms of t?

A. $4t$

B. t^2

C. t^3

D. t^4

E. t^8

238. Last year the price per share of Stock X increased by k percent and the earnings per share of Stock X increased by m percent, where k is greater than m. By what percent did the ratio of price per share to earnings per share increase, in terms of k and m?

A. $\frac{k}{m}\%$

B. $(k-m)\%$

C. $\frac{100(k-m)}{100+k}\%$

D. $\frac{100(k-m)}{100+m}\%$

E. $\frac{100(k-m)}{100+k+m}\%$

239. Of the 300 subjects who participated in an experiment using virtual-reality therapy to reduce their fear of heights, 40 percent experienced sweaty palms, 30 percent experienced vomiting, and 75 percent experienced dizziness. If all of the subjects experienced at least one of these effects and 35 percent of the subjects experienced exactly two of these effects, how many of the subjects experienced only one of these effects?

A. 105

B. 125

C. 130

D. 180

E. 195

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

240. If $m^{-1} = -\frac{1}{3}$, then m^{-2} is equal to

A. -9

B. -3

C. $-\frac{1}{9}$

D. $\frac{1}{9}$

E. 9

241.

DO NOT COPY
filippo.fiz@uptoten.it